

Accessible PDF Example

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1 Overview

This document relies on features introduced in recent versions of L^AT_EX and LuaLaTeX, in particular PDF tagging (via the tagpdf package). As a result, **earlier distributions are not compatible with this template.**

To compile this document successfully, users must update to a current TeX Live distribution and compile using LuaLaTeX:

- **TeX distribution:** TeX Live 2024 (maybe?) or later (TeX Live 2025, 2026 recommended). Instructions are below in Section 2.2.
- **Typesetting Engine:** LuaLaTeX

The choice of editor is flexible. TexShop, TeXworks, VS Code, Overleaf, and command-line workflows are all acceptable, provided they use the correct TeX Live installation and LuaLaTeX. I use TeXShop.

2 Starting Instructions

Fastest: on a Mac, download/update your MacTeX package. This should bring your TeX Live up to 2026.

2.1 Check your TeX Live version (first-time users)

Before converting this document to HTML, it is important to confirm that your computer is using a recent TeX Live installation. This step simply asks LuaLaTeX to report its version number.

How to do this:

1. Open a terminal or command prompt.

- On macOS: Applications → Utilities → Terminal
 - On Windows: Start Menu → Command Prompt or PowerShell
2. In the terminal window, type the following command exactly as shown:

```
lualatex --version
```

3. Press **Enter**.

How to interpret the result.

After running the command, you should see output that includes a line similar to:

```
This is LuaHBTeX, Version 1.22.0 (TeX Live 2025)
```

[Good] If the output mentions **TeX Live 2025** or **TeX Live 2026**, your installation is compatible and you may proceed to Section 3.

[Bad] If the output mentions **TeX Live 2024** (or earlier), or does not find TeX Live, then you need to install a newer version. Proceed to Section 2.2 below.

2.2 Installing or updating TeX Live

TeX Live installs are versioned by year and do not overwrite previous installations. Updating therefore means installing a newer version alongside an existing one.

1. Download the TeX Live installer from TeX Live. Quick download links are here.
2. **macOS (Terminal installation method).** If you are comfortable copying and pasting a few commands, this is a reliable way to install the newest TeX Live.
 - (a) Open the **Terminal** application.
 - (b) Go to your Downloads folder:

```
cd ~/Downloads
```
 - (c) Extract the installer archive:

```
tar -xzf install-tl-unx.tar.gz
```
 - (d) Enter the newly created installer directory:

```
cd install-tl-*
```
 - (e) Start the installer with administrator privileges:

```
sudo ./install-tl
```
 - (f) When prompted, choose **No** (n) when asked whether to import settings from a previous TeX Live installation, then type **I** to start the installation.

3. Run the installer and, if prompted, select the *full scheme*. This is strongly recommended to avoid missing-package issues.
4. After installation (which may take a long time!), verify that the new installation is being used by running the following command from a terminal:

```
lualatex --version
```

The output should indicate TeX Live 2025 (or newer). If it still reports an older version, then your system PATH is likely pointing to an earlier installation.

For example, on macOS with TeX Live 2025 installed, a typical LuaLaTeX path looks like:

```
/usr/local/texlive/2025/bin/universal-darwin/lualatex
```

If you get stuck or are not comfortable using Terminal, it is completely fine to ask for help. One option is to use ChatGPT or similar and start with a prompt like this:

I need to update my TeX Live installation to the newest version so that I can compile accessible PDFs with LuaLaTeX. I was told to run the TeX Live installer and then make sure my PATH points to the new TeX Live. I have no idea what to do. Can you walk me through it step-by-step? I use TeXShop (or whatever).

3 Using your tex editor

Open your tex editor and set the compilation engine to LuaLaTeX, as shown for TeXShop below. No editor-specific features are required by this template.

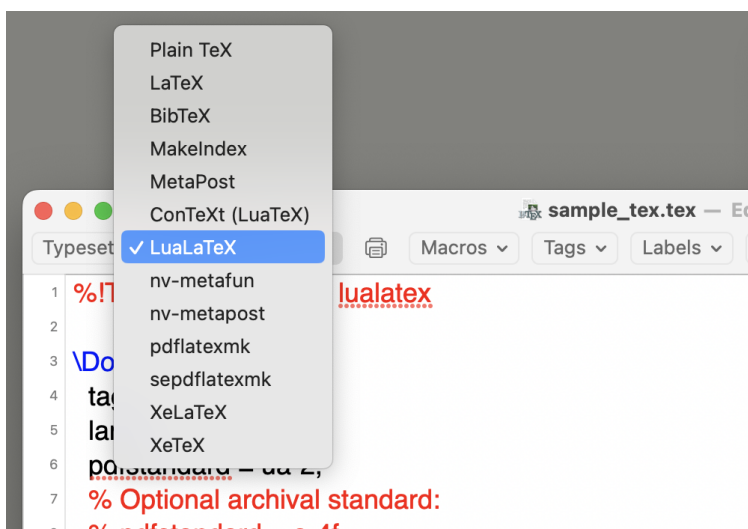


Figure 1: In TeXShop, choose LuaLaTeX from the list just to the right of the Typeset button.

3.1 Mathematical Content

You will write mathematics as you have basically always done in LaTeX. For example,

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\ &= 3x^2.\end{aligned}$$

More complex expressions, displayed equations, and aligned environments are all supported by this workflow, provided they use standard `amsmath` environments.

3.2 Tables

Tables should have clear headers. The first row is treated as a header row via `table/header-rows=1`.

Table 1: Sample table with header row

Topic	Notes
Limits	Use clear language and define notation.
Derivatives	Provide both rules and examples.

Here is how this table was produced:

```
% Leave an empty line before \tagstructbegin

\tagstructbegin{tag=Part}
\begin{table}
\caption{Sample table with header row}
\centering
\begin{tabularx}{\linewidth}{|X|X|}
\hline
\textbf{Topic} & \textbf{Notes} \\ \hline
Limits & Use clear language and define notation. \\ \hline
Derivatives & Provide both rules and examples. \\ \hline
\end{tabularx}
\end{table}
\tagstructend
```

3.3 Figures with alt text

If a figure has a caption, do **not** use `[H]` (from the `float` package), as it can break PDF/UA-2 tagging. Instead, use a standard float placement such as `![htbp]` (or `[t]`, `[b]`, `[p]`) and keep the `\caption` inside the `figure` environment. Include meaningful `alt` text in `\includegraphics`.

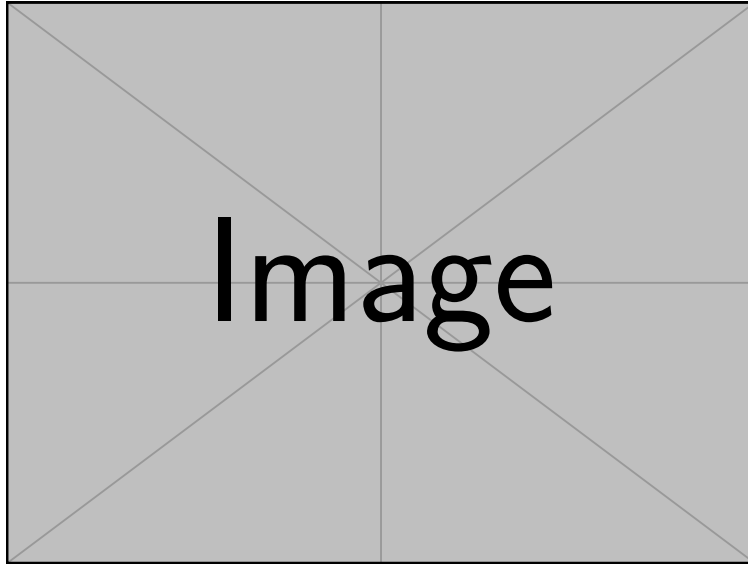


Figure 2: Example figure (replace with your own image).

Here is how this image was produced:

```
% Leave an empty line before \tagstructbegin

\tagstructbegin{tag=Sect}
\begin{figure}
\centering
\includegraphics[width=.6\textwidth,alt={A placeholder example image}]{example-image}
\caption{Example figure (replace with your own image).}
\end{figure}
\tagstructend
```

4 (Optional) Converting the Document to HTML Using Pandoc

Pandoc is a conversion tool that takes the same `.tex` file used to produce the PDF and generates an HTML file. The resulting HTML contains the same section structure, lists, and mathematics as the PDF.

Documents written in HTML are generally more accessible and responsive than PDFs.

4.1 What you need

- Your `.tex` file (for example, this file `sample_tex.tex`)
- Pandoc installed on your computer. You can find the installer on their website. Installation is typically a standard installer with no additional configuration required.

4.2 Step-by-step instructions

1. Locate the folder on your computer that contains your `.tex` file. This is the same folder where the PDF was generated.
2. Open a terminal or command prompt.
 - On macOS: Applications → Utilities → Terminal
 - On Windows: Start Menu → Command Prompt or PowerShell
3. In the terminal, navigate to the folder containing your `.tex` file. (If you are unsure how to do this, tell ChatGPT where your tex file is located and ask it for help directing your terminal to the same location.)
4. Run the following command to convert this file:

```
pandoc sample_tex.tex -o sample_tex.html
```
5. Press **Enter**. Pandoc will create a new file named `sample_tex.html` in the same folder.

Note: this file is called `sample_tex.tex`; replace that with the name of your own `.tex` file.

4.3 What to expect

After running the command:

- No existing files are modified.
- A new HTML file is created in the same location as the original tex.
- Mathematics is rendered using MathJax when viewed in a web browser. This is appropriate for most web-based workflows and provides broad browser support.

If the command completes without error messages, the conversion was successful. The HTML file can be opened in any modern web browser. You can open the HTML file with a text editor to copy the HTML code itself into a new location (e.g. for Wordpress, Canvas, Moodle).

4.4 Notes on accessibility

While this template is designed with PDF accessibility in mind, HTML output accessibility depends on the chosen MathJax configuration, CSS, and downstream hosting environment. Users are encouraged to test the resulting HTML with screen readers and automated accessibility checkers appropriate to their platform.

Questions? Please write bmaults@ncsu.edu.